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In memory searching device

Abstract

An in memory searching device, including multiple first memory cell strings, a controller, and a sensing circuit, is provided. The first memory cell strings are commonly coupled to a first common bit line. Each of the first memory strings includes multiple first data storage layers. The first data storage layers respectively include multiple first memory cell pairs. The first memory cell pairs are respectively coupled to multiple first word line pairs. The controller selects at least one of the first data storage layers to be at least one selected data storage layer, and provides search data to at least one selected word line pair corresponding to the at least one selected data storage layer. The sensing circuit senses a current on the first common bit line to generate a search result.

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Background/Summary

BACKGROUND

Technical Field

(1) The disclosure relates to an in memory searching device, and in particular to an in memory searching device which can improve search efficiency.

Description of Related Art

(2) With the advancement of electronic technology, the development of artificial intelligence is becoming more and more mature. Subsequently, the demand for a large amount of data searching is generated. Accordingly, the in memory searching device is launched. With the increase in the amount of data to be searched due to the development of artificial intelligence, how to apply the in memory searching device to execute data search operations for a large amount of data which requires high search efficiency has become an important topic for persons skilled in the art.

SUMMARY

(3) The disclosure provides an in memory searching device, which can perform layered data search on a memory to improve the efficiency of a data search operation.

(4) An in memory searching device of the disclosure includes multiple first memory cell strings, a controller, and a sensing circuit. The first memory cell strings are commonly coupled to a first common bit line. Each of the first memory cell strings includes multiple first data storage layers. The first data storage layers respectively include multiple first memory cell pairs. The first memory cell pairs are respectively coupled to multiple first word line pairs. The controller is coupled to the first memory cell strings, selects one of the first data storage layers to be at least one selected data storage layer, and provides search data to at least one selected word line pair corresponding to the at least one selected data storage layer. The sensing circuit is coupled to the first common bit line and senses a current on the first common bit line to generate a search result.

(5) Based on the above, the in memory searching device of the disclosure may divide the memory into multiple data storage layers. When performing the data search operation, the data search operation may be performed on one or more layers of the data storage layers, which can speed up the efficiency of data search.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic diagram of an in memory searching device according to an embodiment of the disclosure.

(2) FIG. 2 is a schematic diagram of another implementation manner of the in memory searching device of the embodiment in FIG. 1 of the disclosure.

(3) FIG. 3 is a schematic diagram of another implementation manner of the in memory searching device of the embodiment in FIG. 1 of the disclosure.

(4) FIG. 4A and FIG. 4B are schematic diagrams of a data search operation of an in memory searching device according to an embodiment of the disclosure.

(5) FIG. 5A to FIG. 5C are schematic diagrams of an in memory searching device and an implementation manner of a data search operation according to another embodiment of the disclosure.

(6) FIG. 6A to FIG. 6D are schematic diagrams of multiple execution manners of a data search operation of a selected data storage layer in an in memory searching device according to an embodiment of the disclosure.

DESCRIPTION OF THE EMBODIMENTS

(7) Please refer to FIG. 1. FIG. 1 is a schematic diagram of an in memory searching device according to an embodiment of the disclosure. An in memory searching device **100** includes multiple memory cell strings **110** and **120**, a controller **130**, and a sensor **140**. The memory cell

strings **110** and **120** are commonly coupled to a common bit line BL. The memory cell string **110** is coupled to the common bit line BL through a selection switch SW1, and the memory cell string **120** is coupled to the common bit line BL through a selection switch SW2. The selection switches SW1 and SW2 are respectively controlled by selection signals SSL1 and SSL2 to be turned on or off. The memory cell strings **110** and **120** may include multiple memory cells. Taking the memory cell string **110** as an example, the memory cell string **110** includes multiple memory cells M1 to MZ connected in series. The memory cells M1 to MZ may be NAND flash memory cells.

(8) Each of the memory cell strings **110** and **120** has multiple memory cells coupled in series. The memory cells on the memory cell string **110** are respectively coupled to multiple word lines WL11 to WL1(Z). The memory cells on the memory cell string **120** are respectively coupled to multiple word lines WL21 to WL2(Z). In the embodiment, the memory cells on each of the memory cell strings **110** and **120** may be divided into groups two by two. Two memory cells in a single memory cell string **110** or **120** may form a data storage layer. Also taking the memory cell string **110** as an example, the memory cells M1 and M2 may form a data storage layer, the memory cells M3 and M4 may form another data storage layer, . . . , the memory cells MZ-1 and MZ may form the last data storage layer in the drawing.

(9) In other embodiments of the disclosure, in a single memory cell string, two non-adjacent flash memory cells may also be applied to form a data storage layer. The method shown in the embodiment of FIG. 1 is only an example for illustration and is not intended to limit the scope of the disclosure.

(10) On the other hand, the controller **130** is coupled to the memory cell strings **110** and **120**. When executing a data search operation, the controller **130** may select at least one of the data storage layers in the memory cell strings **110** and **120** as one or more selected data storage layers. For example, in FIG. 1, when executing the data search operation, the controller **130** may select the data storage layer formed by the memory cells M1 and M2 in the memory cell string **110** as a selected data storage layer SL1. Also, the controller **130** selects two memory cells corresponding to the selected data storage layer SL1 in the memory cell string **120** as a selected data storage layer SL2.

(11) In addition, the controller **130** may provide search data SD to a selected word line pair (including the word lines WL11 and WL12) on the selected data storage layer SL1 and a selected word line pair (including the word lines WL21 and WL22) on the selected data storage layer SL2 to perform the data search operation of the search data SD on the selected data storage layers SL1 and SL2. When the search data SD is digital data, the controller **130** may respectively provide multiple bits of the search data SD to the word lines WL11 and WL12 on the selected data storage layer SL1 and the word lines WL21 and WL22 on the selected data storage layer SL2. When the search data SD is analog data, the controller **130** may respectively provide multiple analog component voltages of the search data SD to the word lines WL11, WL12, WL21, and WL22.

(12) The controller **130** of the embodiment of the disclosure may be constructed by applying a digital circuit or may be implemented by applying a memory controller well known to persons skilled in the art, and there is no particular limitation.

(13) In the data search operation, the memory cells in the selected data storage layers SL1 and SL2 may be compared with the search data SD received on the word lines WL11, WL12, WL21, and WL22 according to storage data thereof, so as to generate a search result. In the embodiment, the comparison operation of the memory cells in the selected data storage layers SL1 and SL2 with the search data SD according to the storage data may generate a current on the common bit line BL. In the embodiment, the sensor **140** coupled to the common bit line BL may generate the search result through sensing the current on the common bit line BL. It is worth mentioning that in the embodiment, data similarity represented by the search result is positively correlated with the current on the common bit line BL.

(14) Please refer to FIG. 2 below. FIG. 2 is a schematic diagram of another implementation manner

of the in memory searching device of the embodiment in FIG. 1 of the disclosure. In FIG. 2, two data storage layers in the memory cell string **110** are selected as selected data storage layers **SL1** and **SL3**. Correspondingly, two data storage layers in the memory cell string **120** are selected as selected data storage layers **SL2** and **SL4**. When executing the data search operation, the search data may be synchronously provided to two selected word line pairs (including the word lines **WL11**, **WL12**, **WL13**, and **WL14**) corresponding to the selected data storage layers **SL1** and **SL3** and two selected word line pairs (including the word lines **WL21**, **WL22**, **WL23**, and **WL24**) corresponding to the selected data storage layers **SL2** and **SL4**. In this way, compared with the implementation manner of FIG. 1, the data search operation of data with a larger size may be executed, and the bandwidth of the data search operation can be effectively improved.

(15) It is worth mentioning that in the implementation manner, the selected data storage layers **SL1** and **SL3** may be adjacently disposed data storage layers. In other implementation manners of the disclosure, the selected data storage layers may also be two data storage layers that are not adjacently disposed. The implementation manner of FIG. 2 is only an example for illustration and is not intended to limit the implementation scope of the disclosure.

(16) Please refer to FIG. 3 below. FIG. 3 is a schematic diagram of another implementation manner of the in memory searching device of the embodiment in FIG. 1 of the disclosure. In FIG. 3, all the data storage layers in the memory cell string **110** may be selected at the same time as selected data storage layers **SL1** to **SLA**. Correspondingly, all the data storage layers in the memory cell string **120** may be selected at the same time as selected data storage layers **SL2** to **SLB**.

(17) When executing the data search operation, the search data may be synchronously provided to selected word line pairs (including the word lines **WL11** to **WL1(Z)**) corresponding to the selected data storage layers **SL1** to **SLA** and selected word line pairs (including the word lines **WL21** to **WL2(Z)**) corresponding to the selected data storage layers **SL2** to **SLB**. In this way, compared with the implementation manners of FIG. 1 and FIG. 2, the data search operation of data with a larger size may be executed, and the bandwidth of the data search operation can be effectively improved.

(18) Please refer to FIG. 4A and FIG. 4B below. FIG. 4A and FIG. 4B are schematic diagrams of a data search operation of an in memory searching device according to an embodiment of the disclosure. In FIG. 4A and FIG. 4B, an in memory searching device **400** includes multiple memory cell strings **411** to **41(N)**. The memory cell strings **411** to **41(N)** are respectively coupled to the common bit line **BL** through selection switches **SW1** to **SW(N)**. The selection switches **SW1** to **SW(N)** are respectively controlled by selection signals **SSL1** to **SSLN**. When executing a data search operation, the selection switches **SW1** to **SW(N)** may be respectively turned on at the same time according to the selection signals **SSL1** to **SSLN**.

(19) In the embodiment, as shown in FIG. 4A, in a first interval of performing the data search operation, in the memory cell strings **411** to **41(N)**, a corresponding data storage layer is set as selected data storage layers **SL11** to **SLN1** at the same time. The controller may provide multiple components of the search data to selected word line pairs (including word lines **WL1_1** to **WL1_512** and word lines **WL2_1** to **WL2_512**) on the selected data storage layers **SL11** to **SLN1**. In this way, the in memory searching device **400** may complete the data search operation of a first layer.

(20) Incidentally, in the data search operation, the in memory searching device **400** may provide a pass voltage **Vpass** to multiple word lines of an unselected data storage layer. In addition, each of the memory cell strings **411** to **41(N)** has multiple switches controlled by selection signals **GSL** and **CSL**.

(21) Next, in FIG. 4B, in a second interval of performing the data search operation, in the memory cell strings **411** to **41(N)**, another corresponding data storage layer is set as selected data storage layers **SL12** to **SLN2** at the same time. The controller may provide multiple components of the search data to selected word line pairs (including word lines **WL3_1** to **WL3_512** and word lines **WL4_1** to **WL4_512**) on the selected data storage layers **SL12** to **SLN2**. In this way, the in memory

searching device **400** may complete the data search operation of a second layer.

(22) According to the description of FIG. 4A and FIG. 4B, it can be analogized that the in memory searching device **400** may select each data storage layer in the memory cell strings **411** to **41(N)** one by one according to the time sequence as a selected data storage layer, so as to execute a data search operation accordingly.

(23) It is worth mentioning that in other embodiments of the disclosure, in the same time interval in the data search operation, corresponding to each memory cell string, multiple data storage layers may also be selected as selected data storage layers, and a data search operation is synchronously performed. In this way, the efficiency of the data search operation can be effectively improved.

(24) Please refer to FIG. 5A to FIG. 5C below. FIG. 5A to FIG. 5C are schematic diagrams of an in memory searching device and an implementation manner of a data search operation according to another embodiment of the disclosure. In FIG. 5A, an in memory searching device **500** includes multiple memory cell strings **511** to **5AN** and a sensor and encoder **501**. The memory cell strings **511** to **51N** are commonly coupled to the common bit line **BL1** through selection switches thereof, the memory cell strings **521** to **52N** are commonly coupled to a common bit line **BL2** through selection switches thereof, . . . , the memory cell strings **5A1** to **5AN** are commonly coupled to a common bit line **BLA** through selection switches thereof. In the embodiment, the memory cell strings **511** to **5AN** may be arranged into a cube structure. In the in memory searching device **500**, each of the common bit lines **BL1** to **BLA** extends along a Y-axis direction, each word line extends along an X-axis direction, and multiple memory cells of each of the memory cell strings **511** to **5AN** are coupled in series along a Z-axis direction. In addition, each of the memory cell strings **511** to **5AN** has multiple switches controlled by the selection signals **GSL** and **CSL**.

(25) The sensor and encoder **501** is coupled to the common bit lines **BL1** to **BLA**. A sensor in the sensor and encoder **501** is used to sense a current on the common bit lines **BL1** to **BLA** to generate a search result. An encoder in the sensor and encoder **501** is used to encode the search result to generate an encoded search result.

(26) Incidentally, the sensor in the sensor and encoder **501** may be a sense amplifier and may be implemented by applying a sense amplifier circuit well known to persons skilled in the art. The encoder in the sensor and encoder **501** may be a digital circuit and may be disposed in a page buffer.

(27) When the in memory searching device **500** executes a data search operation, in a single time interval, the in memory searching device **500** may set a data storage layer corresponding to the position of each of the memory cell strings **511** to **5AN** as a selected data storage layer. In the embodiment, the in memory searching device **500** may respectively set the selected data storage layers **SL11** to **SL1N** for the memory cell strings **511** to **51N**, respectively set the selected data storage layers **SL21** to **SL2N** for memory cell strings **521** to **52N**, . . . , and respectively set the selected data storage layers **SLA1** to **SLAN** for the memory cell strings **5A1** to **5AN**.

(28) In the embodiment, based on the mutual correspondence in terms of position, the selected data storage layers **SL11** to **SL1N** are respectively coupled to the corresponding selected word line pairs with the selected data storage layers **SL21** to **SL2N**, and the selected data storage layers **SL21** to **SL2N** are also respectively coupled to the corresponding selected word line pairs with the selected data storage layers **SLA1** to **SLAN**.

(29) On the other hand, in the data search operation, the search data **SD** may be split into multiple parts, which are respectively provided to the selected word line pairs coupled to the selected data storage layers **SL11** to **SLAN**. The selected word line pairs include word lines **WL89_1** to **WL89_512** and word lines **WL90_1** to **WL90_512**. The word lines **WL89_1** to **WL89_512** respectively form multiple word line pairs with the word lines **WL90_1** to **WL90_512**.

(30) In addition, a word line of an unselected data storage layer receives the pass voltage **Vpass**.

(31) In the data search operation of FIG. 5A, the number of the selected data storage layer of each of the memory cell strings **511** to **5AN** is 1 ($n=1$). Moreover, when executing the data search

operation, the in memory searching device **500** may sequentially set a data storage layer of each of the memory cell strings **511** to **5AN** according to the time sequence in different time intervals through the controller to obtain a selected data storage layer, and then execute the data search operation through sending the search data **SD** to the selected word line pairs of the selected data storage layer.

(32) In the implementation manner of FIG. 5B, the in memory searching device **500** may set the number of the selected data storage layers of each of the memory cell strings **511** to **5AN** to 2 ($n=2$) in the same time interval of the data search operation. Also, the data search operation is performed through splitting the search data **SD** into multiple components, which are respectively provided to the selected word line pairs corresponding to the selected data storage layers. In the embodiment, the selected word line pairs include the word lines **WL1_1** to **WL4_1**, **WL1_2** to **WL4_2**, . . . , and **WL1_512** to **WL4_512**.

(33) When executing the data search operation, the in memory searching device **500** may sequentially set two data storage layers of each of the memory cell strings **511** to **5AN** according to the time sequence in different time intervals through the controller to respectively obtain two selected data storage layers, and then execute the data search operation through sending the search data **SD** to the selected word line pairs of the selected data storage layer.

(34) In the implementation manner of FIG. 5C, the in memory searching device **500** may set all the data storage layers of each of the memory cell strings **511** to **5AN** as the selected data storage layers at the same time in the data search operation. The number n of the selected data storage layers is $Z/2$, where Z is the number of memory cells in each of the memory cell strings **511** to **5AN**. Also, the data search operation is performed through splitting the search data **SD** into multiple components, which are respectively provided to the selected word line pairs corresponding to the selected data storage layers. In this way, the data search operation for all storage data of the entire in memory searching device **500** may be completed through a one-time data search operation.

(35) According to the above description, it can be known that the in memory searching device **500** of the disclosure may perform the data search operation through setting the number n of the selected data storage layers according to the requirement of timekeeping, which can effectively improve the efficiency of the data search operation.

(36) Please refer to FIG. 6A to FIG. 6D below. FIG. 6A to FIG. 6D are schematic diagrams of multiple execution manners of a data search operation of a selected data storage layer in an in memory searching device according to an embodiment of the disclosure. In FIG. 6A, a selected data storage layer **610** of the embodiment of the disclosure includes memory cells **MC1** and **MC2**. The memory cells **MC1** and **MC2** may respectively be a single-level cell (SLC), a multi-level cell (MLC), a triple-level cell (TLC), a quad-level cell (QLC), or a storage cell that stores analog data. Word lines of the memory cells **MC1** and **MC2** may respectively receive bias voltages **Vg1** and **Vg2** generated according to the search data, and generate a current **Is1** reflecting the search result on the common bit line **BL1** according to the search result.

(37) In FIG. 6B, the selected data storage layer **610** may have storage data **DAT**. The storage data **DAT** includes a first bit **D1** and a second bit **D2**. The storage data **DAT** may be encoded according to voltage distributions **VT1** to **VT4** of threshold voltages **Vth1** and **Vth2** of the memory cells **MC1** and **MC2**, as shown in the following table.

(38) TABLE-US-00001 TABLE 1 [Vth1 Vth2] DAT = [D1 D2] [VT1 VT4] [0 0] [VT2 VT3] [0 1] [VT3 VT2] [1 0] [VT4 VT1] [1 1] [VT1 VT3] [0 X] (for example, [0 0] or [0 1]) [VT1 VT2] [X 0] (for example, [1 0] or [0 0]) [VT3 VT1] [1 X] (for example, [1 0] or [1 1]) [VT2 VT1] [X 1] (for example, [0 1] or [1 1]) [VT4 VT4] Invalid data [VT1 VT1] Don't care

(39) When executing the data search operation, voltages respectively received by gates of the memory cells **MC1** and **MC2** through the corresponding word lines may be one of voltages **VH1** to **VH4**. The voltages received by the memory cells **MC1** and **MC2** are generated according to the

corresponding search data, wherein taking the two-bit search data as an example, the corresponding relationship between the search data and the voltages VH1 to VH4 is shown in Table 2.

(40) TABLE-US-00002 TABLE 2 Voltage Search data [VH1 VH4] [0 0] [VH2 VH3] [0 1] [VH3 VH2] [1 0] [VH4 VH1] [1 1] [VH4 VH4] Wildcard

(41) When performing the data search operation, the memory cells MC1 and MC2 may generate the current Is1 on the bit line BL1 according to a voltage difference between the voltages received by the gates and storage data thereof.

(42) Next, please refer to FIG. 6C. The voltage distributions VT1 to VT4 respectively being 0V.fwdarw.1V, 1.5V.fwdarw.2.5V, 3V.fwdarw.4V, and 4.5V.fwdarw.5.5V, and the voltages VH1 to VH4 respectively being 5.5V, 7V, 8.5V, and 10V are taken as an example. The peak values of the voltage distributions VT1 to VT4 respectively correspond to the voltages 0.5V, 2V, 3.5V, and 5V. When executing the data search operation, when the search data is [0 0], the bias voltages

(43) Vg1 and Vg2 respectively received by the gates of the memory cells MC1 and MC2 are respectively equal to VH1 and VH4. If the storage data of the memory cells MC1 and MC2 is [0 0] at this time, the threshold voltage Vth1 of the memory cell MC1 is in the voltage distribution VT1, and the threshold voltage Vth2 of the memory cell MC2 is in the voltage distribution VT4. At this time, on the memory cell MC1, a difference value between the bias voltage Vg1 and the threshold voltage Vth1 may be $5.5V - 1V = 4.5V$. Calculated by taking every 1.5V as a level L, the difference value between the bias voltage Vg1 and the threshold voltage Vth1 on the memory cell MC1 may be 3L. On the memory cell MC2, a difference value between the bias voltage Vg2 and the threshold voltage Vth2 may be $10V - 5.5V = 4.5V$. The difference value between the bias voltage Vg2 and the threshold voltage Vth2 on the memory cell MC2 may be 3L.

(44) In other words, under this condition, when the search data matches the storage data, the memory cells MC1 and MC2 may generate the current Is1 corresponding to 3L.

(45) When executing the data search operation, when the search data is [0 0], the bias voltages Vg1 and Vg2 respectively received by the gates of the memory cells MC1 and MC2 are respectively equal to VH1 and VH4. If the storage data of the memory cells MC1 and MC2 is [0 1] at this time, the threshold voltage Vth1 of the memory cell MC1 is in the voltage distribution VT2, and the threshold voltage Vth2 of the memory cell MC2 is in the voltage distribution VT3. The difference value between the bias voltage Vg1(=VH1) and the threshold voltage Vth1(=VT2) of the memory cell MC1 may be 2L. The difference value between the bias voltage Vg2(=VH4) and the threshold voltage Vth2(=VT3) of the memory cell MC2 may be 4L.

(46) Since the memory cells MC1 and MC2 are connected in series, the selected data storage layer 610 may generate the current Is1 corresponding to 2L.

(47) When executing the data search operation, when the search data is [0 0], the bias voltages Vg1 and Vg2 respectively received by the gates of the memory cells MC1 and MC2 are respectively equal to VH1 and VH4. If the storage data of the memory cells MC1 and MC2 is [1 0] at this time, the threshold voltage Vth1 of the memory cell MC1 is in the voltage distribution VT3, and the threshold voltage Vth2 of the memory cell MC2 is in the voltage distribution VT2. The difference value between the bias voltage Vg1(=VH1) and the threshold voltage Vth1(=VT3) of the memory cell MC1 may be 1L. The difference value between the bias voltage Vg2(=VH4) and the threshold voltage Vth2(=VT2) of the memory cell MC2 may be 5L. Since the memory cells MC1 and MC2 are connected in series, the selected data storage layer 610 may generate the current Is1 corresponding to 1L.

(48) When executing the data search operation, when the search data is [0 0], the bias voltages Vg1 and Vg2 respectively received by the gates of the memory cells MC1 and MC2 are respectively equal to VH1 and VH4. If the storage data of the memory cells MC1 and MC2 is [1 1] at this time, the threshold voltage Vth1 of the memory cell MC1 is in the voltage distribution VT4, and the threshold voltage Vth2 of the memory cell MC2 is in the voltage distribution VT1. The difference value between the bias voltage Vg1(=VH1) and the threshold voltage Vth1(=VT4) of the memory

cell MC1) may be 0L. The difference value between the bias voltage $V_{g2}(=V_{H4})$ and the threshold voltage $V_{th2}(=V_{T1})$ of the memory cell MC2 may be 6L. Since the memory cells MC1 and MC2 are connected in series, the selected data storage layer 610 may generate the current I_{s1} corresponding to 0L.

(49) It is not difficult to know from the above description that the size of the current I_{s1} generated by the data storage layer 610 is positively correlated with the similarity between the search data and the storage data.

(50) It is worth mentioning that as long as the number of voltage distributions and the number of bias voltages received by the memory cells MC1 and MC2 are appropriately adjusted, the selected data storage layer 610 of the embodiment of the disclosure may also be used to store storage data of various different levels, and execute the data search operation of the search data of the corresponding level.

(51) In FIG. 6D, the memory cells MC1 and MC2 in the selected data storage layer 610 may also be used to record analog storage data and perform a data search operation for analog search data. The threshold voltage of the memory cell MC1 may be a voltage curve VT_A , and the threshold voltage of the memory cell MC2 may be a voltage curve VT_B . The memory cells MC1 and MC2 may be respectively and independently programmed, so that the threshold voltages respectively correspond to the voltage curves VT_A and VT_B .

(52) In addition, currents I_A and I_B respectively represent channel currents of the memory cells MC1 and MC2.

(53) In the embodiment, the search data may be search voltages V_A and V_B , wherein the search voltages V_A and V_B are related by an expression $V_B = V_{max} + V_{min} - V_A$, where V_{max} and V_{min} are respectively the maximum analog search voltage and the minimum analog search voltage in the data search operation. In the embodiment, the maximum analog search voltage V_{max} and the minimum analog search voltage V_{min} are both constants, such as 9V and 0V respectively.

(54) When the search voltage V_A is equal to 0.7V and the search voltage V_B is equal to 8.3V, the memory cell MC1 does not provide current, but the memory cell MC2 may provide current. When the search voltage V_A is equal to 4.05V and the search voltage V_B is equal to 4.95V, neither the memory cell MC1 nor the memory cell MC2 provides current. When the search voltage V_A is equal to 8.52V and the search voltage V_B is equal to 0.48V, the memory cell MC1 provides current, but the memory cell MC2 does not provide current.

(55) In this way, the size of the current I_{s1} generated by the memory cells MC1 and MC2 in the selected data storage layer 610 may be used as the basis for the degree of similarity between the storage data and the search data in the generated search result. Similarly, the size of the current I_{s1} is positively correlated with the similarity between the storage data and the search data in the search result.

(56) Incidentally, in other embodiments of the disclosure, the voltage curves of the threshold voltage of the memory cell MC1 and the threshold voltage of the memory cell MC2 may also be exchanged. In other words, the threshold voltage of the memory cell MC1 may have the voltage curve VT_B , and the threshold voltage of the memory cell MC2 may have the voltage curve VT_A .

(57) In summary, the in memory searching device of the disclosure sets one or more data storage layers in the memory cell string as one or more selected data storage layers, and provides multiple parts of the search data to the selected word line pairs of one or more selected data storage layers at the same time, so that the data search operation may be synchronously executed for the one or more selected data storage layers. In this way, the rate of the in memory data search operation can be effectively increased to improve the performance of the data search operation.

Claims

1. An in memory searching device, comprising: a plurality of first memory cell strings, commonly coupled to a first common bit line and respectively coupled to a plurality of selection signals, wherein each of the first memory cell strings comprises a plurality of first data storage layers, the first data storage layers respectively comprise a plurality of first memory cell pairs, and the first memory cell pairs are respectively coupled to a plurality of first word line pairs; a controller, coupled to the first memory cell strings, selecting at least one of the first data storage layers of each of the first memory cell strings to be at least one selected data storage layer, and providing search data to at least one selected word line pair corresponding to the at least one selected data storage layer; and a sensing circuit, coupled to the first common bit line and sensing a current on the first common bit line to generate a search result, wherein the controller adjusts a number of the at least one selected data storage layer from 1 to $Z/2$, wherein Z is a number of a plurality of memory cells in each of the first memory cell strings.
2. The in memory searching device according to claim 1, wherein when a number of the at least one selected data storage layer is N , the controller synchronously selects N of the first data storage layers to be the at least one selected data storage layer, where N is a positive integer.
3. The in memory searching device according to claim 2, wherein the controller selects N of the first data storage layers to be the at least one selected data storage layer in a time sequence.
4. The in memory searching device according to claim 1, wherein the controller provides a pass voltage to a plurality of unselected word line pairs of a plurality of unselected data storage layers.
5. The in memory searching device according to claim 1, further comprising: a plurality of second memory cell strings, commonly coupled to a second common bit line, wherein each of the second memory cell strings comprises a plurality of second data storage layers, the second data storage layers respectively correspond to the first data storage layers, the second data storage layers respectively comprise a plurality of second memory cell pairs, and the second memory cell pairs are respectively coupled to the same first word line pairs with the corresponding second memory cell pairs.
6. The in memory searching device according to claim 5, wherein the at least one selected data storage layer comprises each of the selected first data storage layers and each of the corresponding second data storage layers.
7. The in memory searching device according to claim 1, wherein when the search data is digital data, the controller respectively provides a plurality of bits of the search data to at least one selected word line pair corresponding to the at least one selected data storage layer.
8. The in memory searching device according to claim 1, wherein when the search data is analog data, the controller respectively provides a plurality of analog component voltages of the search data to at least one selected word line pair corresponding to the at least one selected data storage layer.
9. The in memory searching device according to claim 1, wherein a data similarity of the search result is positively correlated with the current on the first common bit line.
10. The in memory searching device according to claim 1, wherein a threshold voltage of each memory cell in each of the first memory cell pair is programmed in a first voltage range according to corresponding storage data, the controller sets the search data in a second voltage range, and the first voltage range does not overlap with the second voltage range.
11. The in memory searching device according to claim 10, wherein the storage data is digital data or analog data.
12. The in memory searching device according to claim 1, wherein two memory cells in each of the first memory cell pairs are adjacently or non-adjacently disposed.
13. The in memory searching device according to claim 1, further comprising: an encoder, coupled to the sensing circuit and encoding the search result to generate an encoded search result.
14. The in memory searching device according to claim 13, wherein the encoded search result is

multi-bit digital data or analog voltage.

15. The in memory searching device according to claim 1, wherein when a number of the at least one selected data storage layer is N, the selected data storage layers are adjacently or non-adjacently disposed two by two.
